
Robust Cross-Day Hand-Gesture Recognition from Wrist Surface Electromyography Using a 1D CNN

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<https://github.com/gracemacleodd/APS360-EMG-Gesture-Recognition>

1 Brief Project Description

Surface electromyography (sEMG) records the electrical activity of contracting muscles and is a leading modality for wearable human–computer interfaces, prosthetic control, and rehabilitation. The dominant obstacle to deployment is *cross-day degradation*: a model trained on one recording day loses accuracy on a later day as electrode placement, skin impedance, and muscle state shift between sessions [Kaifosh et al., 2025]. This project asks whether a learned-feature 1D convolutional neural network (CNN) generalizes across recording days better than a traditional feature-engineered support-vector machine (SVM), on 16 static hand/finger gestures from 12-channel wrist sEMG. Deep learning is a natural fit: the raw multichannel signal contains temporal and inter-channel structure that hand-crafted features may discard, so a CNN can learn task-specific, potentially session-invariant filters directly from data.

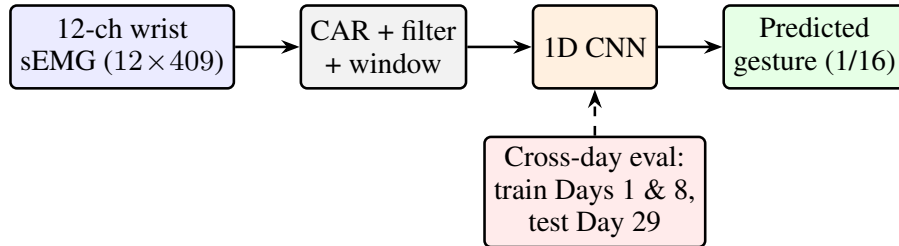


Figure 1: CNN pipeline: wrist sEMG windows in, gesture class out. Cross-day trains on Days 1 and 8, tests on Day 29. The SVM baseline (Sec. 2.2) uses extracted features.

2 Notable Contribution

2.1 Data Processing

Source. We use the open-access GRABMyo dataset [Pradhan et al., 2022, Jiang et al., 2022] (PhysioNet, v1.0.1): 43 participants recorded across three separated days (Days 1, 8, and 29) performing 16 gestures over 7 trials each at 2048 Hz. We restrict to the 12 wrist sEMG channels (dataset columns 18–23 and 26–31; 0-based Python indices 17–22 and 25–30) and use the first 25 participants as a computational development subset selected before evaluating model performance. The complete 43-participant dataset is publicly available on PhysioNet; the final evaluation will scale to all 43.

Cleaning. Each channel is common-average referenced (CAR) across the 12 wrist electrodes, then zero-phase band-pass filtered (4th-order Butterworth, 20–450 Hz via `sosfiltfilt`), and finally 60 Hz notch-filtered ($Q = 30$). No trials were excluded; visual inspection of the filtered signals revealed no flat or saturated channels. Figure 2 shows a raw vs. cleaned channel with a 200 ms analysis window highlighted.

Windowing and statistics. Filtered signals are segmented into ~ 200 ms windows (409 samples, stride 204) at $\sim 50\%$ overlap [Smith et al., 2011], yielding **411,600** total windows balanced across

16 gestures (25,725 per class). Splits are defined at the *trial level before windowing* so overlapping windows never span two splits. The same-day protocol pools Days 1 and 8, splitting by trial: train {1,2,3,5} / val {6} / test {4,7} $\rightarrow$ 156,800 / 39,200 / 78,400 windows. Cross-day trains on trials 1–6 of Days 1 and 8, validates on trial 7, and tests on all of Day 29 $\rightarrow$ 235,200 / 39,200 / 137,200 windows. Per-channel mean and standard deviation are computed on each protocol’s training set only.

Unseen-data plan. For final testing we plan to evaluate on NinaPro DB2 [Atzori et al., 2014], an entirely different sEMG dataset not used during development, serving as an external-domain robustness test with an explicit label mapping to the subset of shared gestures.

Challenges. The full dataset exceeds Colab’s RAM when loaded at once; we implemented a two-pass memory-safe reassembly that preallocates the output array and streams one participant shard at a time.

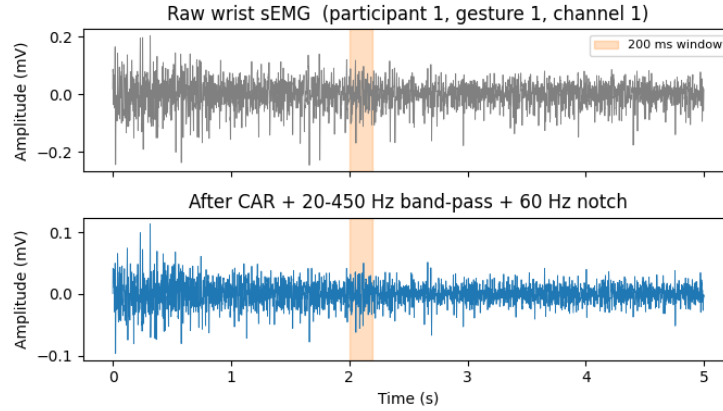


Figure 2: Cleaned training sample: raw (top) vs. filtered (bottom) wrist sEMG, participant 1, gesture 1, channel 1. Orange band = one 200 ms window.

2.2 Baseline Model

The baseline extracts four Hudgins time-domain features [Hudgins et al., 1993]—root-mean-square, mean absolute value, waveform length, and zero crossings (sign-change count, no amplitude threshold)—per channel from each window, giving a 48-dimensional vector. Features are standardized (zero mean, unit variance) using training-set statistics only, then classified by an RBF-kernel SVM with manually selected $C = 10$ and $\gamma = \text{scale}$. Because kernel-SVM training scales super-linearly, the training set is capped at 12,000 windows sampled uniformly at random (seed 0).

Quantitative results. The SVM achieves 66.5% same-day and 51.0% cross-day accuracy (Table 1)—well above the 6.25% chance level—with a cross-day drop of 15.5 percentage points.

Qualitative results. Figure 4a shows the SVM’s cross-day confusion matrix. The off-diagonal spread is broad: the worst confusion is true Thumb Extension (TE) predicted as Thumb–Index Finger Extension (TIFE) at 23.7%, followed by TLFO $\rightarrow$ IMFE at 18.4%—errors that may reflect overlapping forearm-muscle activation patterns between gestures.

Challenges. Capping training at 12,000 windows is an acknowledged confound when comparing absolute accuracy with the CNN. We treat each model’s within-model cross-day drop as a protocol-level robustness indicator rather than a perfectly controlled domain-shift estimate.

2.3 Primary Model

The 1D CNN (Figure 3) treats each window as a 12-channel, length-409 time series. It stacks three convolutional blocks—each Conv1D (kernel 5, padding 2), batch normalization, ReLU, max-pool by 2—widening channels $12 \rightarrow 32 \rightarrow 64 \rightarrow 128$. Global average pooling collapses the temporal axis, dropout ($p=0.3$) regularizes, and two FC layers ($128 \rightarrow 64 \rightarrow 16$) produce logits. The network has **63,088 trainable parameters**. Training uses Adam (lr 10^{-3} , weight decay 10^{-4}), cross-entropy

loss, batch size 256, mixed-precision on GPU, ReduceLR0nPlateau scheduling, and early stopping (patience 8) on validation macro-F1.



Figure 3: 1D CNN (63,088 params). Each Conv block: Conv1D(5)→BN→ReLU→MaxPool2.

Quantitative results. The CNN achieves 85.2% same-day and 74.1% cross-day accuracy (Table 1), outperforming the SVM in every setting. Its cross-day drop is 11.1 percentage points versus the SVM’s 15.5, consistent with learned convolutional features being more robust to the day-to-day domain shift. The comparison does not fully isolate architecture from training-set size (the SVM is capped at 12,000 windows), so the within-model drop serves as a protocol-level robustness indicator. All metrics are computed on the pooled test set across participants.

Table 1: Test accuracy and macro-F1 (25 participants, 16 gestures). 411,600 is the total processed dataset; per-protocol test sizes are 78,400 (same-day) and 137,200 (cross-day). “Drop” = same-day – cross-day accuracy.

Model	Same-day		Cross-day		Cross-day drop
	Acc	F1	Acc	F1	
RBFSVM (Hudgins)	0.665	0.665	0.510	0.508	0.155
1D CNN	0.852	0.852	0.741	0.740	0.111

Qualitative results. Figure 4 compares the cross-day confusion matrices (row-normalized, shared 0–100% colour scale). The CNN’s diagonal is darker and its off-diagonal mass more contained. Its top confusions are TLFO→IMFE (19.0%) and IMFE→TLFO (15.0%)—a symmetric pair that may reflect similar finger-opposition mechanics—plus LP→HC (14.5%), where the lateral pinch and hand close may share overlapping grip-muscle activation. The SVM’s worst confusion (TE→TIFE, 23.7%) is both larger and reflects a broader difficulty distinguishing extension gestures. Notably, cross-day validation macro-F1 plateaus near 0.85–0.87 on Days 1 and 8, while held-out Day-29 test macro-F1 is 0.740—directly illustrating the cross-day domain shift.

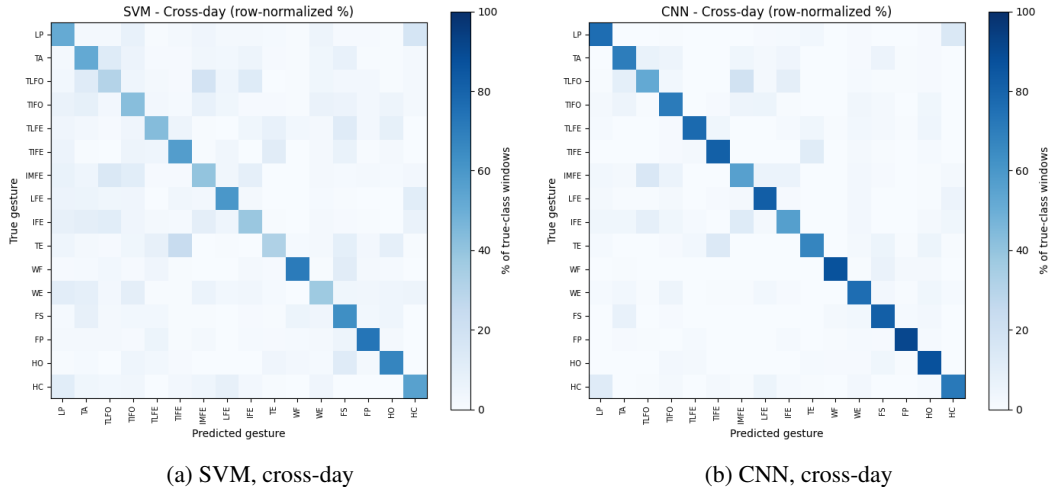


Figure 4: Cross-day confusion matrices (row-normalized %, shared 0–100% scale). The CNN’s errors are more contained along the diagonal.

Challenges and feasibility. Validation is drawn from Days 1 and 8 and does not reflect the Day-29 shift; the true cross-day performance is the held-out test accuracy. The cross-day CNN reached the 60-epoch cap without early stopping; whether additional training helps will be examined. Training takes ~5–7 min per CNN on Colab Pro, so the remaining work is tractable.

References

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